

WHEN FORGETTING LOOKS LIKE IMPROVEMENT: METRIC MASKING IN STREAMING DIARIZER ADAPTATION AND THE PRICE OF REHEARSAL

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ABSTRACT

Small-data adaptation can improve speech detection while degrading speaker attribution. We study this discrepancy in a released streaming diarizer adapted on 7.5 h of two-party conversation and evaluated across six corpora. Adaptation substantially improves in-domain diarization performance and transfers to an independent corpus. However, this improvement is not consistent across evaluation scenarios as the additional confusion is mainly associated with impaired temporal identity consistency rather than speaker-count errors. A local-remapping diagnostic reveals different patterns of identity degradation across corpora, indicating that adaptation may alter how streaming models maintain speaker assignments over time. Rehearsal reduces the observed degradation but reduces the cross-domain transfer performance. These results highlight the need to jointly evaluate detection accuracy, identity consistency, and retention behavior when adapting streaming diarization systems.

Index Terms— Streaming speaker diarization, domain adaptation, capability retention, rehearsal, temporal consistency

1. INTRODUCTION

Streaming speaker diarization must detect speaker activity and maintain consistent identities as audio arrives [1]. When combining with a Streaming Sortformer it addresses online identity tracking through a speaker cache [2]. Fine-tuning such a pretrained model for a target domain raises an important question of **what improves, and what retains?** While Aggregate diarization error rate (DER) is commonly used to indicate performance, the questions remains unclear as gains sometimes hide other attributes such as increased speaker confusion.

Prior work has investigated continual self-supervised domain adaptation [3]. Our focus is on how these adaptation changes detection, speaker assignment, and temporal consistency across domains [4]. We conduct a diagnostic study of

Streaming Sortformer v2, comparing the released checkpoint with target-only adaptation and mixed-data rehearsal variants across six corpora. We first decompose DER, then equalize framewise active-output counts to compare assignment under identical scored detection errors. Short-window speaker remapping relaxes recording-wide correspondence without changing predictions. History interventions further probe sensitivity to streaming state, while the rehearsal comparison examines recovery and sacrifices.

Our contribution is threefold. First, we demonstrate a gap between score-level retention and speaker-assignment retention; second, we uncover history-sensitive consistency costs beyond short-window confusion, revealing a temporal dimension of adaptation that aggregate scores obscure; and third, the tested rehearsal recipe redistributes retention and transfer, as aggregate recovery comes with lost transfer gains, and near-baseline DER can still conceal assignment degradation.

2. EXPERIMENTAL PROTOCOL

2.1. Models, adaptation, and evaluation data

The released `diar_streaming_sortformer_4spk-v2` model [5] (**Stock S**) has four speaker slots ordered by first appearance [6] and produces activity scores at 12.5 Hz. **Expert (E)** fine-tunes all 117 M parameters for 24 epochs using AdamW [7], learning rate 10^{-5} , batches of four 60-s crops, oversampling factor two, gradient clipping at 5.0, and an auxiliary Dice-loss weight of 0.2 in NeMo [8]. We use the final-epoch checkpoint. **Generalist (G)** adds rehearsal [9, 10] 5. All models use mono channel 0 and the same released inference implementation at the 1.04-s latency configuration.

The model is fine-tuned on MMCSG dataset [11], its ’s official evaluation partition is re-split into 153 training sessions (7.5 h) and 36 speaker-disjoint holdout sessions (1.8 h). The current training and holdout sets are disjoint, but the holdout participated in earlier recipe development. Eight sessions were in the earlier validation split and 28 in its training split. Aside from in-domain evaluation, we also evaluate PriMock57 [12] and EasyCom [13]. Development corpora including MSDWild [14] few.val, AliMeeting-Test [15]’s two-

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speaker subset, and AMI [16] single-distant-microphone test were used for informed rehearsal development.

2.2. Metric scope and operating points

For frame t , let r_t and h_t count reference speakers and active output slots. On scoring frames $\mathcal{T}$, count-based global Miss and false alarm are

$$\begin{aligned} M(\mathcal{T}) &= \frac{100 \sum_{t \in \mathcal{T}} (r_t - h_t)_+}{\sum_{t \in \mathcal{T}} r_t}, \\ F(\mathcal{T}) &= \frac{100 \sum_{t \in \mathcal{T}} (h_t - r_t)_+}{\sum_{t \in \mathcal{T}} r_t}. \end{aligned} \quad (1)$$

Within-overlap Miss (ov-Miss) restricts $\mathcal{T}$ to $r_t \geq 2$. Both miss measures ignore speaker identity. For AMI and EasyCom, count metrics and threshold calibration additionally restrict $\mathcal{T}$ to $r_t \leq 2$. Standard DER uses the full continuous reference [17]. Count Miss and DER miss are not interchangeable. A separate diagnostic recomputes DER within the count mask, with its own optimal mapping.

For each corpus, we select Stock’s threshold nearest a nominal count-FA target of 2.62%, then select each adapted model’s threshold nearest Stock’s achieved FA on a 0.005 grid. Table 1 reports actual count FA. All thresholds use labels from the scored corpus, including the two recipe-independent corpora. These are controlled diagnostic operating points. Sensitivity analyses use 2.0% and 3.5% targets and grouped two-fold calibration on AMI and EasyCom.

2.3. Uncertainty and replication

We use nine speaker-connected groups for MMCSG, ten parent sessions for EasyCom, and recording units for PriMock57 (57), MSDWild (274), and AliMeeting-2 (8), with 20,000 resamples. For AMI, we group recordings into four meeting series and enumerate all 256 bootstrap sequences. Intervals are conditional on the checkpoints, thresholds, and grouping; they do not resolve unverified participant dependencies or training uncertainty and are not adjusted for multiple comparisons. Three Expert seeds are evaluated on AMI, EasyCom, and PriMock57.

3. ADAPTATION IMPROVES DETECTION WHILE ALTERING IDENTITY RETENTION

3.1. Adaptation with heterogeneous retention

Expert reduces MMCSG ov-Miss from 33.75% to 22.13% and DER from 15.57% to 7.82% (Table 1). The paired DER change is -7.74 pp [$-9.73, -5.35$]. On recipe-independent PriMock57, ov-Miss improves by 15.43 pp and DER by 10.72 pp [$-11.94, -9.65$]. These gains show substantial adaptation headroom. Retention is domain-dependent: DER increases

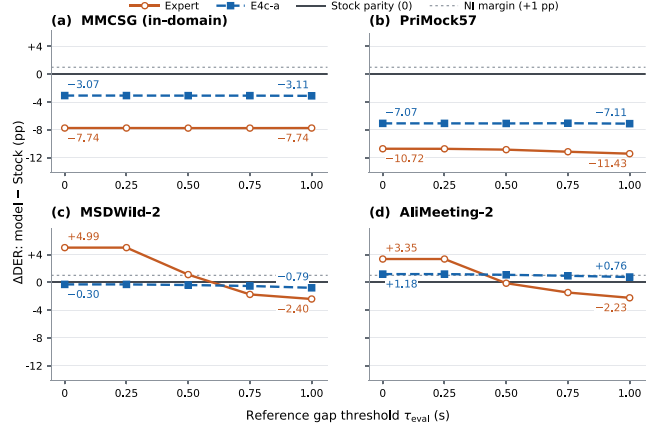


Fig. 1. Changing of convention-aligned labels lower the DER on MSDWild-2 and AliMeeting-2 datasets but remain unchanged for in-domain and PriMock57.

by 4.99 pp on MSDWild and 3.35 pp on AliMeeting-2, predominantly through missed detection, while AMI and EasyCom exhibit a different error composition.

3.2. Detection gains can conceal identity degradation

On AMI, full-scope DER miss falls from 24.90% to 22.87% and DER FA from 2.59% to 2.49%, yet confusion increases from 7.75% to 21.46%.

EasyCom provides independent corroboration of the identity deficit. Expert lowers gMiss by 2.97 pp [$-5.62, -0.04$] while increasing confusion by 3.52 pp [$+2.62, +4.47$]. Overall DER changes by $+0.79$ pp [$-1.53, +3.39$]. This distinction shows that detection gains partly compensate for an identity loss that the decomposed evaluation detects. Scoring DER within the same $r_t \leq 2$ mask preserves the point-estimate reversal on both corpora (AMI $+11.03$ pp; EasyCom $+0.67$ pp). Scoring scopes alone do not explain the phenomenon.

AliMeeting-2 illustrates a complementary effect as its 24.06-pp ov-Miss increase corresponds to only 3.35 pp of DER degradation. Decomposing the 3.42-pp increase in global count Miss attributes 2.56 pp (74.9%) to overlap and 0.86 pp to non-overlap. Overlap-restricted scoring exposes a concentrated weakness that whole-recording averaging reduces in magnitude.

3.3. Replication and calibration sensitivity

The replications support the direction of the principal domain contrasts without treating seeds as additional evaluation samples. Across three Expert seeds, dataset exhibits decreased gMiss while increased confusion and DER. At nominal count-FA targets of 2.0%, 2.62%, and 3.5%, Expert has lower gMiss but higher DER on both AMI and EasyCom. The DER increases range from 11.31 to 11.59 pp and from 0.27 to 1.52

Table 1. Adaptation and retention under matched count FA. S/E/G denote Stock/Expert/Generalist.

Corpus	Role	Model	Count FA	gMiss	ov-Miss	DER miss	DER FA	Confusion	DER
MMCSG (36)	ID	S	2.612	12.36	33.75	12.37	2.61	0.59	15.57
		E	2.618	4.93	22.13	4.95	2.62	0.25	7.82
		G	2.620	4.88	24.02	4.90	2.62	0.31	7.83
PriMock57 (57)	I	S	2.582	15.86	41.77	15.87	2.59	0.68	19.15
		E	2.593	5.43	26.34	5.46	2.62	0.34	8.42
		G	2.613	15.36	37.83	15.37	2.63	1.40	19.40
EasyCom (223)	I	S	2.577	31.28	50.88	32.80	2.46	7.34	42.60
		E	2.562	28.31	49.26	30.09	2.44	10.86	43.39
		G	2.526	27.16	49.76	28.89	2.42	8.56	39.87
MSDWild (274)	D	S	2.608	17.18	40.91	17.20	2.64	1.88	21.72
		E	2.721	22.71	45.35	22.72	2.74	1.25	26.71
		G	2.593	15.10	34.28	15.12	2.63	2.10	19.85
AliMeeting-2 (8)	D	S	2.631	3.07	9.93	3.11	2.66	0.12	5.90
		E	2.621	6.49	33.99	6.51	2.63	0.11	9.25
		G	2.617	2.77	8.62	2.81	2.65	0.13	5.59
AMI (16)	D	S	2.627	23.70	35.91	24.90	2.59	7.75	35.23
		E	2.629	20.15	40.79	22.87	2.49	21.46	46.82
		G	2.654	15.89	34.44	17.98	2.54	8.58	29.10

pp, respectively. Grouped two-fold calibration preserves this direction in pooled held-out results for all three distinct balanced AMI partitions and five EasyCom partitions. The checks establish persistence over the examined operating points, not threshold-invariant effect sizes or significance in every fold.

4. DIAGNOSING THE IDENTITY DEFICIT

4.1. Local assignment and recording-wide consistency

We retain predictions and compare recording-wide speaker mapping with independent optimal mappings over non-overlapping windows of length W . Let C be pooled confusion with one optimal mapping per recording and L_W its locally mapped counterpart, using identical time support and reference speaker-time. Define: $C = L_W + G_W$, $G_W = C - L_W \geq 0$.

G_W measures the penalty associated with that constraint; L_W is the confusion remaining after relaxation. This is an oracle scoring diagnostic, related to local/global evaluations [18]. It can expose time-varying slot assignments or slot reuse without distinguishing their internal causes.

On AMI, the added confusion is dominated by $\Delta G_{30} = 12.12$ pp, compared with $\Delta L_{30} = 1.59$ pp (Table 2). Across three Expert seeds, 30-s remapping removes 88–91% of the added confusion. The large mapping penalty persists at 10 and 60 s and after excluding 30-s windows containing only one reference identity. At 30 s, EasyCom retains 61–67% of its added confusion locally across seeds. This attribution is scale-dependent, for example, 10-s remapping removes 71–74% added EasyCom confusion and 33–39% at 30 s. Its clips are at most 60 s, so $G_{60} = 0$ follows from the file length.

Table 2. Confusion decomposition for Expert seed 0. Changes against Stock are pp with paired 95% intervals.

Corpus	ΔC	ΔL_{30}	ΔG_{30}
Original predictions			
AMI	+13.71	+1.59	+12.12
EasyCom	+3.52	+2.16	+1.36
PriMock57	-0.34	-0.05	-0.29
Reference-count-assisted decoding			
AMI	+14.91	+0.45	+14.46
EasyCom	+3.75	+1.73	+2.02
PriMock57	-1.65	-1.26	-0.39

4.2. Controlling speaker count

To test whether identity degradation is explained solely by count errors, we select the top- r_t output slots using the reference concurrent speaker count. Original and assisted predictions are rescored on identical eligible support (Table 2). Assistance still leaves an AMI confusion increase of 14.91 pp from G_{30} and EasyCom retains increases in both components. AMI’s 10-s local confusion is lower than Stock across seeds, despite higher recording-wide confusion. We freeze each system’s original mapping and compare times when both predict the correct count and found that conditional confusion increases by 19.10 pp on AMI and 4.04 pp on EasyCom (58.89% and 50.66%). Instantaneous count errors alone therefore do not explain the identity deficits.

4.3. Speaker returns as a temporal probe

On AMI, we compare the last and first up to 5 s of a speaker’s preceding and returning reference turns. Both sides must contain at least 1 s where the reference and both systems have

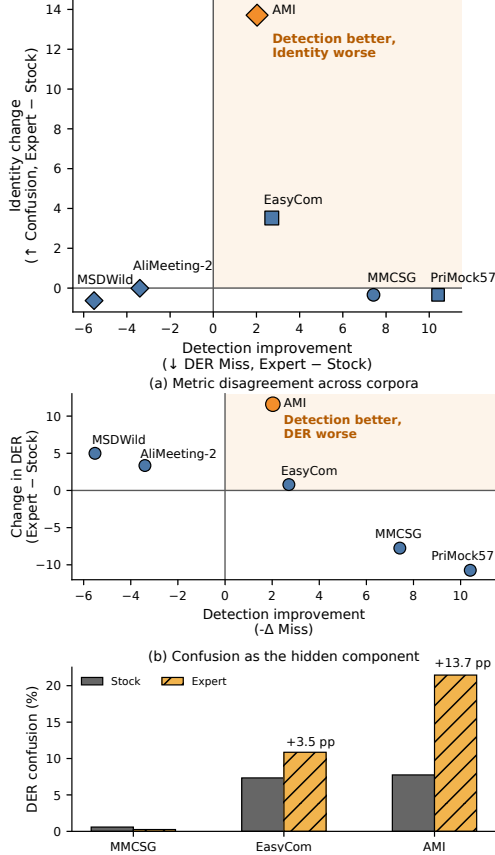


Fig. 2. Adaptation changes diarization capabilities beyond aggregate DER. Detection improvements can coexist with increased speaker confusion, which is hidden when only aggregate DER is considered.

exactly one active speaker. With recording-wide mappings fixed, we measure the pre-to-post change in Expert-minus-Stock confusion. The long-minus-short contrast, for speaker inactivity of at least 30 s versus [2, 10) s, is positive for all three seeds: 0.98, 1.01, and 9.98 pp. Only 20–23 long-gap events qualify from 636 candidates, supports differ across model pairs with various interval length. This suggests a conditional association with return timing, rather than establishing a general long-gap failure or a causal cache mechanism.

5. REHEARSAL GAINS AND THEIR COST

Generalist combines target-domain crops with AMI-train, AliMeeting-Eval, and MSDWild few.train. The final rule aims for equal rehearsal contributions and a 1:1 rehearsal-to-target ratio. Integer oversampling yields 92, 96, and 80 rehearsal crops versus 306 target crops per epoch and uses 3,432 optimizers (an 88.2% increase). This comparison measures the complete rehearsal recipe under approximately matched nominal target exposure.

Generalist reduces DER relative to Expert on AMI, MSD-

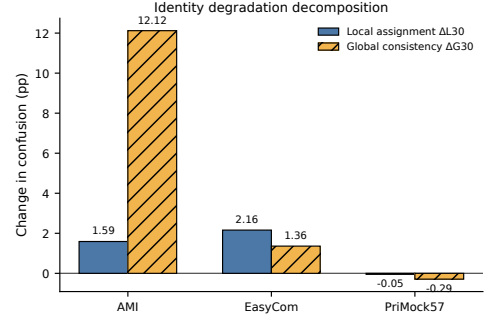


Fig. 3. Decomposition of increased confusion into local assignment error and recording-wide consistency penalty.

Wild, AliMeeting-2, and EasyCom. AMI’s confusion reduction is primarily a decrease in G_{30} with -12.40 pp, while the local change is -0.49 pp. On EasyCom, both components improve, but confusion remains 1.22 pp above Stock. Rehearsal therefore mitigates the identity deficit without uniformly restoring the original error composition. On PriMock57 the Generalist gives up 11.49 pp of ov-Miss and 10.98 pp of DER relative to Expert, eliminating the DER transfer benefit. The measured retention gains consequently come with a transfer cost and do not establish universal non-degradation.

5.1. Pause annotation and partial recovery

Without retraining, we merge the same speaker reference gaps and repeat matched-FA calibration. τ_{eval} from 0 to 1 reverses Expert-Stock DER on MSDWild-2 (+4.99 to -2.4 pp) and AliMeeting-2 (+3.35 to -2.23 pp), see Figure 1.

Exploratory training-label reconstruction using word timestamps and calibrated padding retains a 7.07 pp PriMock gain and 1 pp for MSDWild-2, with AliMeeting-2 still degrades of 1.18 pp. These results support annotation conventions as a partial contributor to degradation, with orpus-dependent repair costs.

6. LIMITATIONS AND CONCLUSION

We showed that small-data adaptation can substantially improve in-domain diarization and enable cross-domain transfer, yet detection gains coexist with degraded speaker identity consistency. Evaluation showed that speaker-count errors alone cannot explain this deficit, while local remapping distinguished local assignment errors from recording-wide consistency losses. Rehearsal mitigated degradation and largely preserved in-domain DER gains, but sacrificed transfer benefits and left residual identity degradation, demonstrating that DER recovery does not necessarily imply recovery of speaker-assignment behavior. Future work will extend the analysis to other streaming architectures, adaptations, and rehearsal strategies for balance retention and transfer.

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